Circuits Updates - June 2024

We report a number of developing ideas on the Anthropic interpretability team, which might be of interest t researchers working actively in this space. Some of these are emerging strands of research where we expect publish more on in the coming months. Others are minor points we wish to share, since we're unlikely to ev paper about them.

We'd ask you to treat these results like those of a colleague sharing some thoughts or preliminary experime few minutes at a lab meeting, rather than a mature paper.

New Posts

- Comparing TopK and Gated SAEs to standard SAEs
- Research By Other Groups

Comparing TopK and Gated SAEs to standard SAEs

Hoagy Cunningham, Tom Conerly

Background

There have been a few different papers in recent months proposing alternative approaches to training SAEs works point to big wins being available in the Average Number of Active Features/Mean Squared Error (hen L0 and MSE) tradeoff space, primarily due to alleviating the shrinkage issue in which the L1 penalty makes tl optimal-loss level of feature activation lower than the 'true' activation leading to less accurate reconstructio given level of sparsity.

These papers include:

- Gated Sparse Autoencoders (Rajamanoharan et al., GoogleDeepmind)
- Using a TopK penalty instead of an L1 (Gao et al, OpenAI)
- Finetuning a subset of the parameters after the main SAE training phase (Wright and Sharkey, MATS

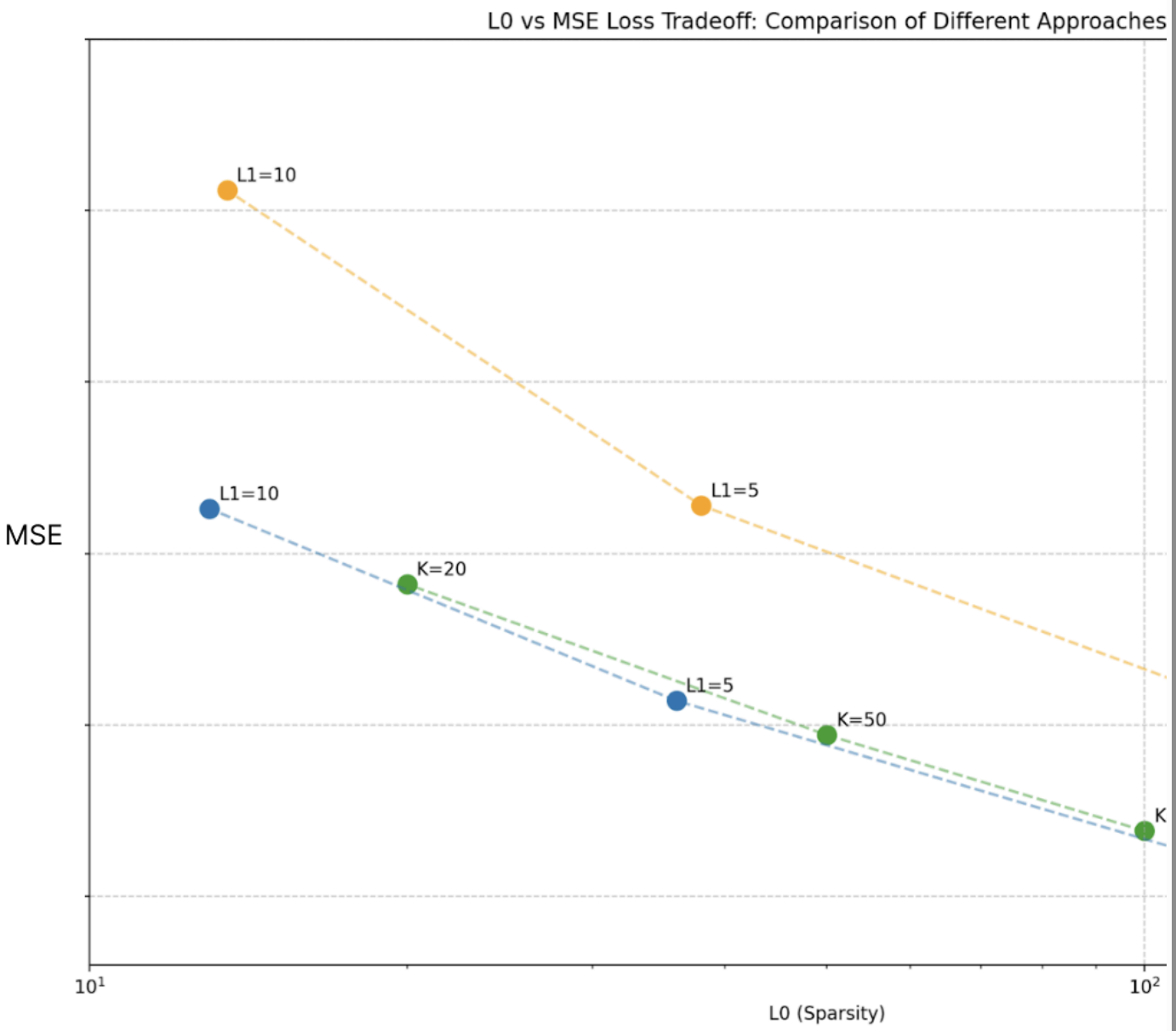
There was also a post in our February Update - Using a Tanh penalty. (Jermyn et al) - which found that repla L1 penalty with a Tanh function improved the L0/MSE tradeoff, but they also found that this resulted in sign less interpretable features. Rajamanoharan et al compare the Gated Sparse Autoencoders (Gated SAEs) to st SAE features and find no significant difference in the degree of interpretability, while Gao et al compare mea interpretability between TopK hyperparameters, but do not compare to standard or Gated SAEs.

We are therefore keen to check that we can replicate these improvements in L0/MSE, and also, especially fo TopK SAEs, that these changes aren't accompanied by regressions in the degree of interpretability of the fou features.

L0/MSE Results

We ran a replication of some of these claims and confirmed that we see a significant benefit from shifting fr standard, L1 penalized SAEs, to either Gated SAEs or TopK SAEs, with the performance of the latter two bein We used tied weight initialization between the encoder and decoder for both models, and used the unconst decoder norm (see April 2024 Circuits Update) on the Gated SAE but not the TopK SAE.[1]

Our basic results on L0/MSE are as follows are shown in the graph below - note that for TopK the L0 is by de equal to the value of K).



Notably, the amount of optimization that's gone into training the TopK and Gated SAEs is lower than that g the standard SAEs. For example, we haven't implemented the variant of Ghost Grads that Gao et al used to dead features in their recent paper. Despite this, it's clear that there's a very significant improvement in (L0, here, as previously demonstrated by Gao et al and by Rajamanoharan et al.

Density

We can also see significant differences in the distribution of densities between the different approaches. The runs have more features at the upper end of the density range than the Gated SAE, even when the Gated SA higher L0. Both TopK and Gated SAEs display a greater number of dense features than the standard approac

Run Type	L0	Features > 1e-2	Features > 1e-3
TopK - K=50	50	261	8076
TopK - K=20	20	54	2855
Gated SAE	37	8	4998
Vanilla SAE	38	13	360

We can also look at the average level of feature activation, and here the two TopK runs stand out even more greater degree of variation in the average activation, with gated SAEs being intermediate between the TopK standard runs.

Blinded evaluations

It's clear that TopK represents a Pareto improvement over the standard SAE on the L0/MSE tradeoff, but ulti declare it an improvement over the existing baseline we need to confirm that the quality of the features fou comparable or better.

There are two broad ways to evaluate the SAE - we can evaluate the patterns of when features fire, and ther try to evaluate the quality of the learned basis and how well this basis allows us to understand the semantic vector space.

To evaluate the first we evaluated the degree of monosemanticity of the firing patterns of the features. We f a manual blind test of the features, in which the set of top-activating contexts were displayed, alongside the seven of the activation intervals as displayed in feature visualizations like in Scaling Monosemanticity.

The lower buckets were not displayed as both Gated SAEs and TopK SAEs have noticeably different patterns activations which would have removed the blinding, and it's unclear that the monosemanticity of these low activations is an important metric for these features.

Activation Consistency

5: Clear pattern with no deviating examples

4: Clear pattern with one or two deviating examples

3: Clear overall pattern but quite a few examples not fitting that pattern

2: Broad consistent theme but lacking structure

1: No discernible pattern

Complexity

5: Rich feature firing on diverse contexts with an interesting unifying theme, e.g. "feelings of togetherness"

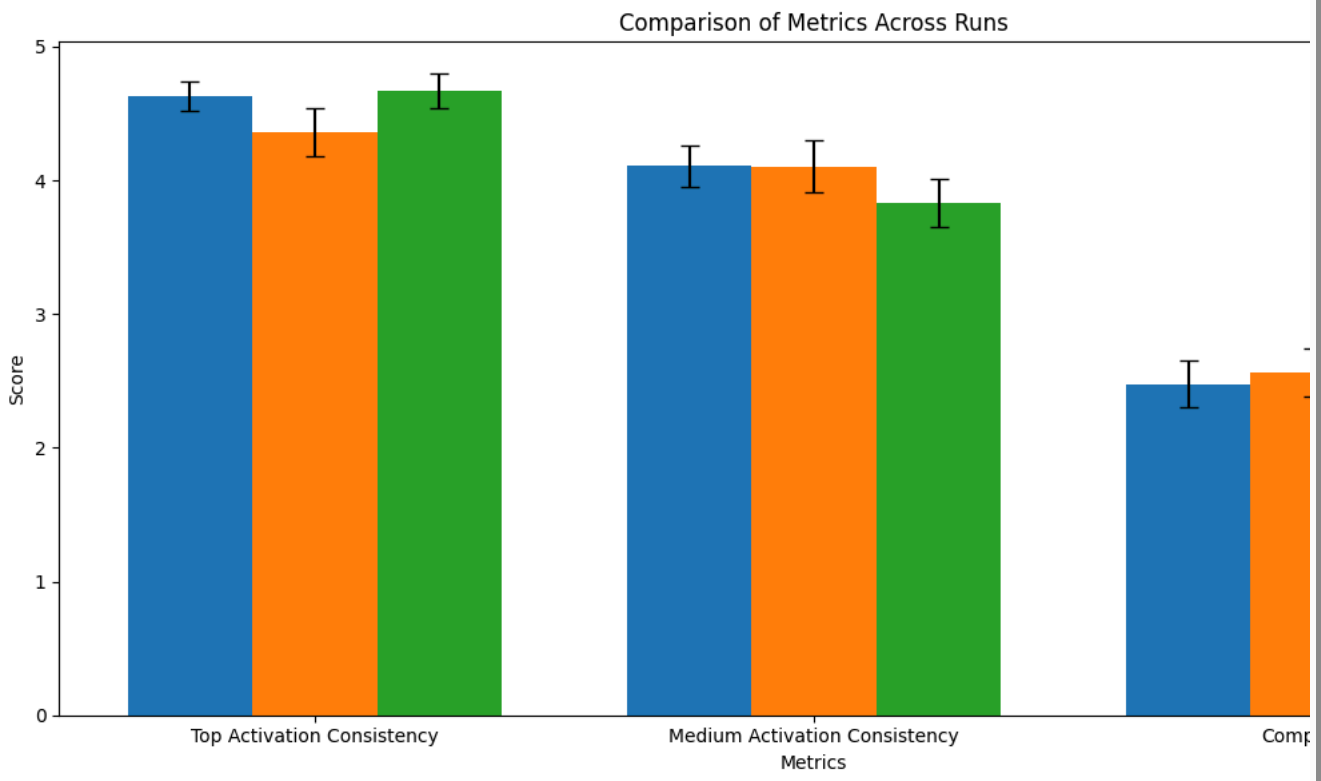
4: Feature relating to high-level semantic structure, e.g. “return statements in code”

3: Moderate complexity, such as a phrase, category, or tracking sentence structure e.g. “website URLs”

2: Single word or token feature but including multiple languages or spelling e.g. “mentions of dog”

1: Single token feature e.g. “ the token ‘(’”

We manually scored 100 features using this rubric without access to which run the features were from. It see there’s no discernible difference between the different runs at this scale of analysis and any difference will b



As mentioned above, TopK runs exhibit higher density features than are found in comparable standard SAEs isn’t necessarily a problem but they have a higher than average impact on the reconstruction so it’s importa check that they are also interpretable. To this end we also looked specifically at the highest density features with K=20, n_features=256k on an 18 layer model.

Firstly, there’s a lot of variation in the extent of high density features. Some runs with higher values of K had features in the 20-50% region with no clear explanation and these are more worrying. The regime we care n about however is that with a very large number of features and high sparsity, and with the improvements in we can target lower L0 values than with standard SAEs.

- The highest density feature fired about 9% of the time, there were 7 features firing more than once tokens and several dozen firing more than once in 100 tokens - so overall this didn’t seem to be a si issue.
- The highest density features were often related to tokens, firing either on them or passages of text s after them.

- The lower density features included features for common words such as “the” and “of”, as well as lar like Italian and Russian and pre-20th century English.
- There were a few unintelligible features, often connected to code, in the 1-2% range but this numbe unexpected, and given that the rest of the features made clear sense, it can’t be ruled out that these were playing a computational role that wasn’t clear just from their contexts.

In contrast, in a K=100 run, there we see dozens of high density features which activate more than 1 in ever tokens and which seem largely uninterpretable, and which would be a worrying sign, but given the positive lower K values, mostly just suggests that we should aim for greater levels of sparsity.

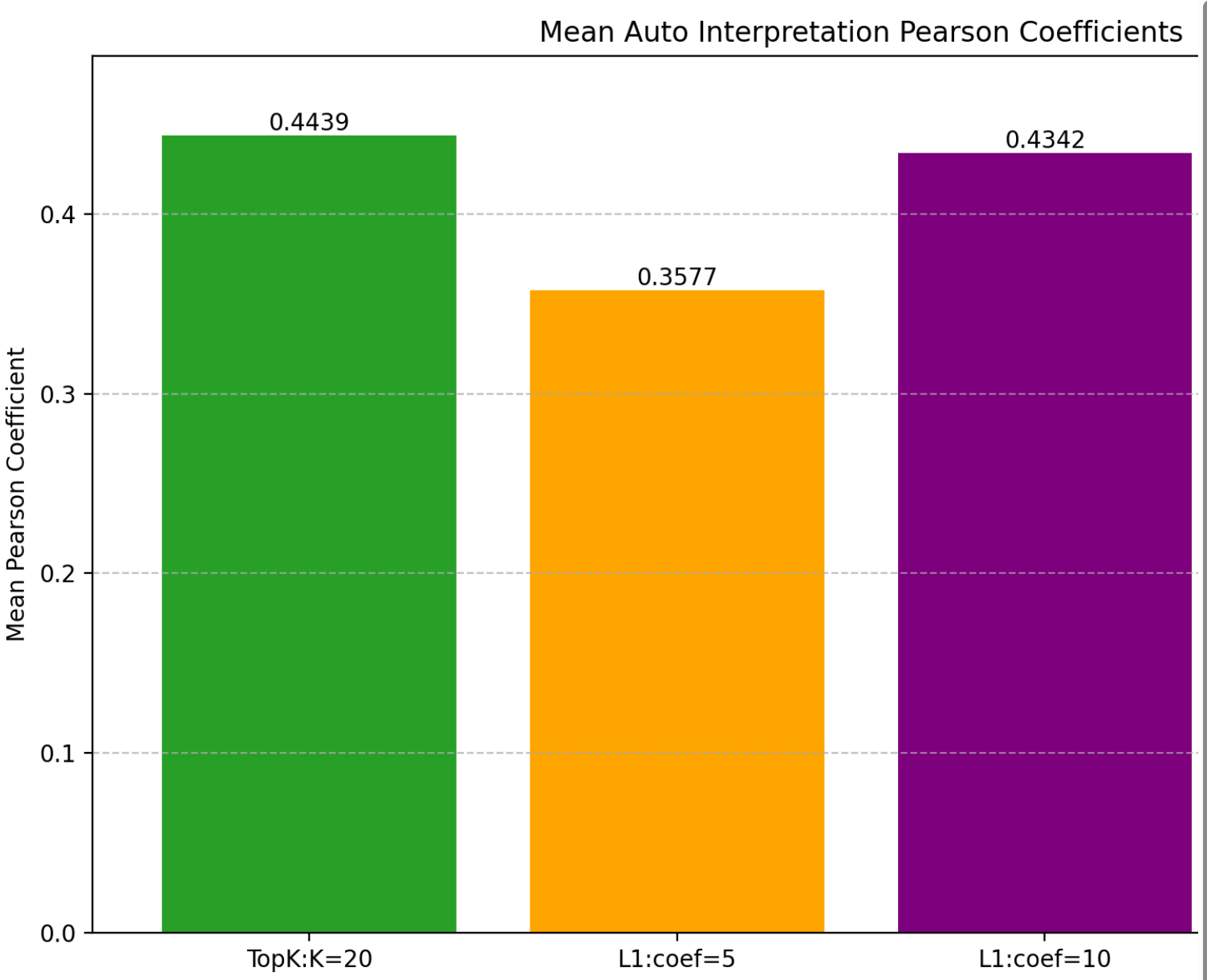
It therefore seems that these high density features are a common outcome of shrinkage-reduction methods pathological at high values of L0, but also these methods allow for much lower L0 at comparable levels of M and when using these low L0 runs the features are of at least comparable quality to those found with standa

The conclusion is that these high density features don’t seem to be pathological, and this also suggests that finding from Anthropic that using Tanh(x) instead of the L1 norm as the sparsity penalty, which also leads to improvements on L0/MSE along with high density features, may not have been as pathological as previously at least with high number of learned features in high sparsity regimes.

Clerp

We also ran 200-250 features from each of four runs through Clerp, our automatic interpretation system, wh generates predicted feature activations based on the model-generated feature explanations and scores eac by the correlation between the real and predicted feature activations.

We found that the scores for the L1 = 5 Gated SAE and K=20 TopK SAE both score as well as the L1=10 star which has far worse MSE loss and lower L0. These results are clearly consistent with both Gated SAEs and To a step forward in terms of interpretability, though it may be that by having many fewer lower activations, th are easier while the upper and middle activation ranges are very similar, which is consistent with the manua above.



More detail on Clerp can be found in Towards Monosemanticity.

Future Work

The above results give us confidence that both Gated SAEs and TopK SAEs are strong alternatives to standa with little downside risk and the potential to be a meaningful improvement. However, it's still difficult to knc whether the basis found by an SAE is better or worse. For example, an approach which solves or reduces shr may significantly improve the MSE loss while the underlying feature basis remains unchanged. Gao et al try measure the quality of the basis more directly by finding the quality of the best possible probe for certain b datasets that use only a direction. This seems a principled approach but it's not clear which kinds of dataset should expect features to discover clean probes for.

Ultimately SAEs need to be judged on whether they provide additional insight into how the model works - c use it to debug model issues? For steering? For finding circuits? For understanding the impact of fine-tuning improve robustness? It's clear that the evaluations we have at the moment don't get to the heart of what we about and we're excited to work on that and for future work from others which fills this gap.

Research by Other Groups

Brian Chen, Rodrigo Luger, Adam Pearce, Josh Batson, Hoagy Cunningham

Scaling SAEs to GPT-4

Scaling and Evaluating Sparse Autoencoders by Leo Gao et al from OpenAI present an alternative method o sparse autoencoders in which they use a TopK activation function instead of the ReLU used in the original S By removing the L1 pressure it removes the issue of shrinkage, which is where the optimal loss is achieved b feature having a smaller activation value than the true value. It also allows us to set the L0 norm directly whi much more understandable hyperparameter that gets at the thing we're interested in much more directly th penalty, and instead more directly penalizes the sparsity that we are interested in. They show empirically th L0-MSE curve which is commonly used to evaluate SAEs that it's a clear improvement over the vanilla SAE a similar to the Gated SAE of Rajamanoharan et al. In their SAE training they find two key tools for reducing th number of dead features - firstly to initialize the encoder as the transpose of the decoder, and secondly usir variant of the 'ghost grads' approach in which features which have not fired for some fixed period of time a encouraged to model the reconstruction error with an auxiliary loss using the highest-activating dead featu

They also show a pair of novel methods of trying to evaluate the models. Firstly they take a number of exist classification datasets and, for each SAE, find the accuracy of the best possible linear probe that can be built the probe direction is aligned with one of the SAE features. They find that in experiments on GPT-2 the prob at a minimum for SAEs with L0 in the range 64–128, and decreases with the number of features. The change probe quality over training are rather modest, and it's unclear to what extent the probes are the kind of thin one should expect to exist as single features, but the ability to evaluate SAEs on terms that are totally extern technique is exciting and is complementary to the approach of Makelov et al as discussed below. They also their SAEs by the sparsity of the effect of ablating the feature on the logits, here finding that the greater deg sparsity is found in the L0 range of 128–256.

See the section in this month's update on comparing TopK, GatedSAEs with standard SAEs for more informa which replicates Gao et al's claim that the TopK SAEs are a step forward in the L0/MSE tradeoff. We are glad other frontier AI labs sharing the goal of deeply understanding the mechanisms underlying the capability of models and are grateful for their contributions to this line of research.

Evaluating Interpretability Methods

As we've previously discussed, one major obstacle in the development of sparse autoencoders is the lack of "ground truth" objective. The metrics we optimize are only proxies for interpretability. For example, we abar our attempts to train autoencoders with a tanh penalty after finding that the resulting features were harder interpret, even though they improved L0 and loss recovered. We are therefore excited for the development methods for evaluating SAEs.

In RAVEL: Evaluating Interpretability Methods on Disentangling Language Model Representations, Huang et propose a benchmark that tests an interpretability method's ability to isolate attributes and causally interve them.

For example: Paris is in Europe, and people in Paris speak French. A good interpretability method should be disentangle the model's representation of these two aspects of Paris. To evaluate this, one can obtain the m representation of a similar prompt featuring, say, Tokyo, and then replace only the part of the representatio method suggests is about the continent. The interpretability method succeeds if, after this activation patchi intervention, the model reports that Paris is in Asia, but continues to report that people in Paris speak Frenc

The authors of RAVEL test several methods on this benchmark and find that the best method is a variation c Distributed Alignment Search, which learns a linear subspace of a model that has the desired effect when in on. Although the authors primarily evaluate RAVEL on supervised methods, we are most excited about usin as a benchmark for unsupervised methods, as we believe that many behaviors of interest in LLMs may not b captured in a dataset that can be used to train disentanglement. One potential limitation of RAVEL is that it the case that some concepts cannot be independently manipulated in the model – French, for example, is a language whose origins tie it to Europe, and manipulating its location separately from its language may not entirely possible without damaging the model. While the dependence of the properties may provide an upp on the score, relative comparisons between methods on the same model may still be made.

In Towards Principled Evaluations of Sparse Autoencoders for Interpretability and Control, which outlines m concerns with evaluating SAEs we share, Makelov et al. develop a different evaluation method on the indire identification (IOI) task, a popular task in interpretability introduced by Wang et al., 2023. In this task, the m completes a prompt similar to "When Mary and John went to the store, John gave a book to" with the name The authors list three relevant attributes that each IOI prompt has: the subject (S; John), the indirect object (and their relative positions (IO-S-S, as opposed to S-IO-S). They then learn supervised dictionaries for the a attributes, which form a "ground truth", as well as two kinds of unsupervised dictionaries, "task SAEs" train the IOI dataset and "full-distribution SAEs" trained on OpenWebText, a general text corpus. Finally, they eva these dictionaries with the following criteria:

- Sufficiency and Necessity: If activations are replaced with their reconstructions according to an SAE, model still do the task? Conversely, if activations are replaced with their error term (plus the mean) a to an SAE, does the model fail to do the task?
- Sparse Controllability: For a given model and prompt, can one find a sparse set of activations to edi changes the output in an expected way (e.g., complete our example prompt with "John" instead of " The authors formulate this as an NP-hard discrete optimization problem and approximate it with a g algorithm.
- Interpretability: Can some of the features be understood as some simple/interpretable predicate ov "ground truth" attributes?

The first two criteria can be evaluated without reference to the "ground truth" attributes. Only the last metri depends on the specific choice of attributes, as well as additional human judgment in deciding what kinds c predicates are simple. For example, the authors include a predicate about the typical gender of a name in th which they find reflected in a few learned features. A priori, any specific language model may not use featur corresponding to these specific attributes to solve the IOI task, but the empirical results suggest that the st model, GPT2-small, does.

The authors verify that the supervised dictionaries perform well according to all three criteria, but the unsup dictionaries fall short, with the task SAEs outperforming the full-distribution SAEs. This sets a concrete bench that dictionary learning methods can aspire to improve on. They also report on several phenomena:

- The relevant features in the task SAEs and full-distribution SAEs are quantitatively similar. Several fea the task SAEs also seem to generalize well to the broader distribution.
- Across many hyperparameter choices, SAEs learn features for most IO names, but very few S names. authors term this "feature occlusion" and argue that this is caused by the IO features having larger magnitude.
- Task SAEs often split the binary "relative position attribute" (IO-S-S versus S-IO-S) into multiple featu are hard to interpret. The authors term this "feature over-splitting", but caution that this does not ne indicate an optimization failure in the SAE, since the binary attribute was chosen by human judgmer may not actually be part of the best decomposition of the activations. In support of the idea that the features are real, the authors find that the learned split features across different random seeds are m similar than chance. On the other hand, the authors train some toy models on synthetic activations b two "ground truth" features plus Gaussian noise, and show that these models prefer to learn many r two features.

This work is a step towards evaluating the quality of an overall representation given by an SAE, as opposed quality of an individual feature or its correspondence to a hypothesized probe. Since a model may organize differently than we do, such an ensemble approach to evaluating features is valuable.

Multidimensional Features

Not All Language Model Features Are Linear challenges the *linear representation hypothesis* of LLMs, in whic features represent one-dimensional lines (directions) in some latent space, by exploring the existence of multidimensional feature representations. The authors focus on *statistically irreducible* multidimensional fea features that cannot be decomposed into statistically independent features or into a mixture of non co-occu features.

The two most interesting examples of these multidimensional features are the two-dimensional spaces in w of the week and months of the year are represented; the authors find these features in both Llama 3 8B and 7B using an unsupervised clustering strategy. In the study, the authors focus on the specific task of modular arithmetic with days and months (e.g., "two days from Monday is…" and "four months from November is…") investigate whether these circular representations are utilized for computation, i.e., whether there is a causa relationship between the representation and the output. Through activation patching intervention experime authors demonstrate that circular representations are indeed causally involved in solving these tasks, and th representation is irreducible: the model internally represents these features using more than one dimension Specifically, the authors demonstrate that information is encoded in the *angle* along the day-of-the-week o of-the-year circles. This is best demonstrated by their off-distribution interventions, in which they find that t consistently predicts the correct output token (day or month) regardless of the position of the patched poin circular subspace (e.g., whether it is inside or outside the unit circle) as long as its *angular* position is the sar

The authors further isolate the computation to primarily the MLP layers of the model. While attention heads involved in copying the information needed for the computation to the final token position, intervening on attention heads had relatively small effects compared to intervening on the MLP layers.

Interestingly, the study did not find other readily interpretable multidimensional features. While this could b limitations of the SAE clustering method used to identify these features, it could also mean these irreducible multidimensional features are rare, and that most concepts can be represented as linear combinations of or dimensional features.

[1] The error bars are probably too confident as the activation consistency measures are quite heavy tailed c end and a couple of unintelligible features can have a large impact.

Footnotes

1. To get this to work we had to change L_aux to not freeze the decoder. With the decoder frozen during training the decoder norm kept sh and r_mag kept increasing. [↩]